Homework Project 1

1 Reinforcement Learning with Policy Gradient

Problem 1 Consider a robot moving on a two dimensional plane with obstacles on it. The goal of the robot is to move from an initial position to some desired end position while avoiding obstacles. The initial position of the robot is an obstacle-free position. The plane is subdivided in a finite number of positions (square cells) and extends in the z and y directions with 10 times 5 cells, as shown in Figure 1. At every position, except for an end position, the robot will perform one step forward on the z axis, and simultaneously may choose to make one step along the y axis upwards, downwards, or stays at the current level. In exact, at every position (y, z), $y \in \{1, 2, 3, 4, 5\}$, $z \in \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$,

$$(y, z) \mapsto (\tilde{y}, z + 1), \ \tilde{y} \in \{y - 1, y, y + 1\} \cap \{1, 2, 3, 4, 5\},$$

as shown in Figure 1, and moreover, at the end positions (y, z), $y \in \{1, 2, 3, 4, 5\}$, $z = 10$, there can be no more movement on any axis. The desired end positions are $(y_f, z_f) \in \{(2, 10), (4, 10)\}$, and they are marked with an "X" symbol on Figure 1. A green cells depicts an obstacle-free position, while a red cell depicts a position with an obstacle. The robot may pass through a cell with an obstacle, however, it should be penalized (in terms of obtained reward) for doing so. Moreover, it should not move needlessly along the y axis, i.e. every movement on the y axis should be penalized, however not as much as for going to an obstacle cell. Finally, moving outside the defined two dimensional space is not permitted.

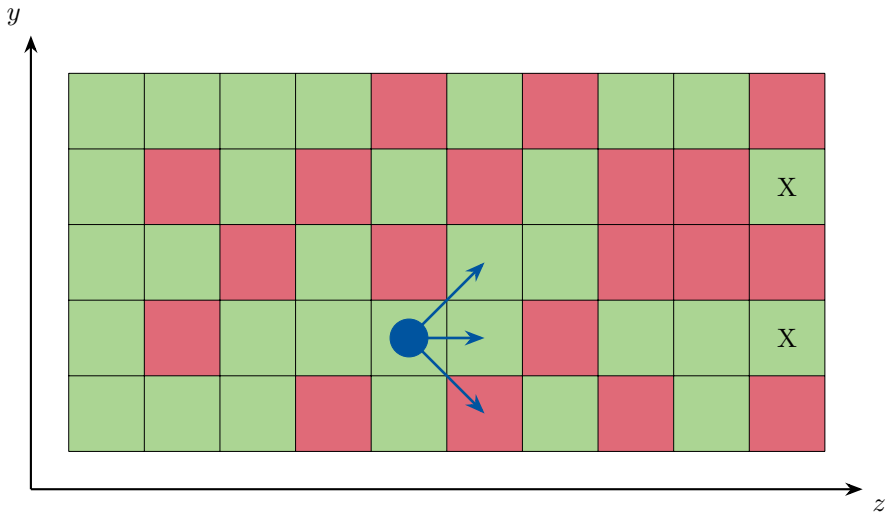


Figure 1: A robot (blue dot) in the 2-dimensional space, with possible next states. A green cell indicates no obstacle while a red cell indicates an obstacle. The symbol X indicates a desired end position.

The first task is to model the robot in the two dimensional space as a Markov Decision Process (MDP). In particular, model

1.a) the state space $\mathbb{X}$,

1.b) the action space $\mathbb{U}$,

1.c) the reward set $\mathcal{R}$, and

1.d) the transition-reward kernel $\mathbf{q}$:

- provide the system dynamics, i.e. the state transitions, and
- provide a reward (regret) function, i.e. the penalty that each transition incurs.

Problem 2 In the following, we analyze the iterative optimization algorithms which we later use to find an optimal policy. Consider some MDP $(\mathbb{X}, \mathbb{U}, \mathcal{R}, \mathbf{q})$, for which both the state and action spaces are discrete. The transition kernel $\mathbf{q}$ of the MDP determines the probability of transitioning from one state (given an input, i.e. action) to a next state and the reward for the transition. The probability of choosing an action at one state is determined by the control policy, and the goal of this problem is to optimize the control policy, i.e. to find the action which provides the "best" possible reward for the transition. In the following, we denote by $\mathbb{X} \times \mathbb{U}$ the set of all state-action pairs (s, ν), where $s \in \mathbb{X}$ and $\nu \in \mathbb{U}$. For clarity, we set $m = |\mathbb{X}|$ and $p = |\mathbb{U}|$, where $|\cdot|$ denotes the cardinality of the set, i.e. the number of distinct elements in it. In exact, the sets $\mathbb{X}$ and $\mathbb{U}$ are given as

$$\mathbb{X} := \{s_1, s_2, \ldots, s_m\}, \qquad \mathbb{U} := \{\nu_1, \nu_2, \ldots, \nu_p\}.$$

To parameterize the to-be-determined control policy, we introduce the parameterization $\theta \in \mathbb{R}^{|\mathbb{X}\times\mathbb{U}|}$, where $|\mathbb{X} \times \mathbb{U}|$ denotes the cardinality of $\mathbb{X} \times \mathbb{U}$, i.e. the number of distinct state-action pairs possible. The vector θ is initialized (chosen) by the user, and then it takes the following form

$$\theta := \begin{bmatrix} \theta(s_1, \nu_1) & \theta(s_1, \nu_2) & \ldots & \theta(s_1, \nu_p) & \theta(s_2, \nu_1) & \theta(s_2, \nu_2) & \ldots & \theta(s_m, \nu_p) \end{bmatrix}^\top.$$

The (stochastic) control policy $\pi : \mathbb{R}^{|\mathbb{X}\times\mathbb{U}|} \times \mathbb{U} \times \mathbb{X} \to [0, 1]$ represents the probability that at some state $s_i \in \mathbb{X}$ the action $\nu_j \in \mathbb{U}$ will be sampled, i.e. taken. For simplicity, we introduce the abbreviation $\pi^\theta(\nu_j; s_i) := \pi(\theta, \nu_j; s_i)$. In this problem, we define the control policy to be the softmax function, i.e. for all $(s_i, \nu_j) \in \mathbb{X} \times \mathbb{U}$

$$\pi^\theta(\nu_j; s_i) = \frac{e^{\theta(s_i,\nu_j)}}{\sum\limits_{\nu_k \in \mathbb{U}} e^{\theta(s_i,\nu_k)}}$$

The softmax function is a $\mathcal{C}^1$ map, (strictly) positive, and satisfies the condition $\sum\limits_{\nu_k \in \mathbb{U}} \pi^\theta(\nu_k; s_i) = 1$ for any $s_i \in \mathbb{X}$. These are exactly the properties needed to define a parameterized discrete probability measure (i.e., a Gibbs distribution) for the policy.

In this project, we consider a finite sequence of states (visited) and actions (taken), i.e., $(x_{0:T}, u_{0:T-1}, r_{1:T})$, and refer to it as a sample trajectory, or simply an episode of length T (horizon T). Notice that for every transition from state $s_i \in \mathbb{X}$ to $s_i^+ \in \mathbb{X}$ if the action $\nu_j \in \mathbb{U}$ is sampled, an (incremental) reward r_k is also obtained, where $1 \leq k \leq T$, which is determined by the transition-reward kernel $\mathbf{q}$. Therefore, for every sample trajectory, we have the reward trajectory $r_{1:T}$. Note that for every sample trajectory there exists no initial reward r_0, as there exists no action (e.g. u_{-1}) which can be penalized. Similarly, once the final state is visited, we do not transition to a further state. Typically, we therefore consider $r_0 = 0$ and $u_T = 0$. Throughout this problem, we denote a sample trajectory of length $T \in \mathbb{N}$ with initial condition $x_0 \in \mathbb{X}$ as

$$\omega_T(x_0) := (x_0, u_0, x_1, u_1, r_1, \ldots, x_T, r_T).$$

It represents a reasonable objective to try to maximize the obtained rewards for all possible trajectories. Let (X_k, U_k, R_k) denote the state, action and reward random variables respectively of the MDP with the parameterized policy π^θ, as described above, for $k = 0$ to $k = T$. Therefore, we represent the problem by $X_{0:T}, U_{0:T-1}$ and $R_{1:T}$, which are finite sequences of random variables, instead of the state, action and reward sequences of the current trajectory, in order to find the optimal policy for all possible trajectories. In particular, given an initial condition $x_0 \in \mathbb{X}$ we aim to maximize the sum of the incremental rewards, i.e. we aim to maximize

$$V^{\pi^\theta}(x_0) := \mathbf{E}\left(\sum_{k=1}^{T} \mathbf{E}\left[R_k | U_{0:k-1}, X_0 = x_0\right]\right) =: \mathbf{E}^{\pi^\theta, x_0}\left(\sum_{k=1}^{T} R_k\right),$$
$$U_i \sim \pi^\theta(\cdot; x_i), \quad i \in \{0, 1, \ldots, T-1\} \quad ,$$

with respect to θ.

To find the optimal policy, it is necessary to optimize the parameterization of the control policy θ w.r.t. some objective function for any initial condition $x_0 \in \mathbb{X}$. Let $J : \mathbb{R}^{|\mathbb{X}\times\mathbb{U}|} \to \mathbb{R}$ be

the objective function. For a particular initial state $x_0 \in \mathbb{X}$, we define the objective function as $J_{x_0}(\theta) := V^{\pi^\theta}(x_0)$, and to account for all possible initial conditions we moreover define $J(\theta) := \sum_{x_0 \in \mathbb{X}} J_{x_0}(\theta) p(X_0 = x_0) = \mathbf{E}\left(J_{x_0}(\theta)\right)$, where $p(X_0 = x_0) = \mu(x_0)$ denotes the probability of the initial condition being x_0.

To maximize the value function we use the gradient ascent iterative optimization method, i.e.

$$\theta_{i+1} = \theta_i + \eta \nabla_\theta J(\theta_i) = \theta_i + \eta \mathbf{E}\left(\nabla_\theta J_{x_0}(\theta_i)\right),$$

where $\eta > 0$ is the step size and $i \geq 0$ is the iteration step. The gradient of $J_{x_0}(\theta_i)$ at any $i \geq 0$ can be determined with the help of the policy gradient theorem:

$$\nabla_\theta J_{x_0}(\theta_i) = \mathbf{E}^{\pi^\theta, x_0}\left(\sum_{k=1}^{T} R_k \sum_{l=0}^{T-1} \nabla_\theta \log \pi^{\theta_i}(U_l; X_l)\right).$$

As $\pi^\theta : \mathbb{R}^{|\mathbb{X}\times\mathbb{U}|} \times \mathbb{U} \times \mathbb{X} \to [0,1]$ and is a $\mathcal{C}^1$ map, it follows that $\nabla_\theta \log \pi^\theta : \mathbb{U} \times \mathbb{X} \to \mathbb{R}^{|\mathbb{X}\times\mathbb{U}|}$. In the following we provide the basic implementation of the Policy Gradient (PG) algorithm:

input : A MDP$(\mathbb{X}, \mathbb{U}, \mathcal{R}, \mathbf{q})$, a differentiable policy π^θ, $\theta_0 \in \mathbb{R}^{|\mathbb{X}\times\mathbb{U}|}$, number of update steps $M \in \mathbb{N}$, (sufficiently large) number of training episodes $N \in \mathbb{N}$, episode length $T \in \mathbb{N}$, step size $\eta > 0$
output : (Sub)optimal parameterization $\theta^* = \theta_M$
for *all* $x_0 \in \mathbb{X}$ **do**
  for $i = 0 : M-1$ **do**
    $g_T \leftarrow 0$; /* $g_T \in \mathbb{R}^{|\mathbb{X}\times\mathbb{U}|}$ */
    for $j = 1 : N$ **do**
      Generate (sample) episode $\omega_T(x_0) := (x_0, u_0, x_1, u_1, r_1, \ldots, x_T, u_T, r_T)$ with control policy π^{θ_i};
      $g_T \leftarrow g_T + \sum_{k=1}^{T} r_k \sum_{l=0}^{T-1} \nabla_\theta \log \pi^{\theta_i}(u_l; x_l)$;
    end
    $\theta_{i+1} = \theta_i + \frac{\eta}{N} g_T$;
  end
end

Algorithm 1: Policy Gradient Algorithm

The Policy Gradient requires many sample trajectories to perform a single update of the parameterization of the control policy θ. Indeed, it is necessary to do so as only convergence in expectation may be proven, and therefore the sample-mean estimate of sufficiently many episode information is used, i.e. $\frac{1}{N} R_T$.

In order to reduce the number of episodes needed to find the optimal policy for the MDP, we present in the following an alternative implementation of the Policy Gradient, which we refer to as the Stochastic Policy Gradient (SPG):

input : A MDP$(\mathbb{X}, \mathbb{U}, \mathcal{R}, \mathbf{q})$, a differentiable policy π^θ, $\theta_0 \in \mathbb{R}^{|\mathbb{X}\times\mathbb{U}|}$, number of update steps $M \in \mathbb{N}$, number of episodes $T \in \mathbb{N}$, step size $\eta > 0$
output : (Sub)optimal parameterization $\theta^* = \theta_M$
for ***all*** $x_0 \in \mathbb{X}$ **do**
  for $i = 0 : M - 1$ **do**
    $g_T \leftarrow 0$; /* $g_T \in \mathbb{R}^{|\mathbb{X}\times\mathbb{U}|}$ */
    Generate (sample) episode $\omega_T(x_0) := (x_0, u_0, x_1, u_1, r_1, \ldots, x_T, u_T, r_T)$ with control policy π^{θ_i};
    $g_T \leftarrow \sum_{k=1}^{T} r_k \sum_{l=0}^{T-1} \nabla_\theta \log \pi^{\theta_i}(u_l; x_l)$;
    $\theta_{i+1} = \theta_i + \eta g_T$;
  end
end

Algorithm 2: Stochastic Policy Gradient Algorithm

For the above problem setup:

2.a) Provide arguments as to why the SPG could be a viable alternative to the "basic" PG,

2.b) Discuss the motivation to use the SPG, analogous to the motivation to use stochastic gradient descent (SGD) instead of gradient descent (GD), and

2.c) Provide arguments as to why SPG may be used instead of PG, and what conditions must be satisfied to have convergence guarantees for the SPG.

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Problem 3 In this problem, we aim to simulate a PG optimization method in order to find an optimal control policy for a MDP. To this end, consider the setup as in Problem 1. For the given MDP with a fixed initial parameterization θ_0 of the control policy π^θ and some sufficiently large number $N \in \mathbb{N}$, implement the Policy Gradient (both the "basic" and stochastic) to find the optimal policy π^{θ^*} and answer the following questions:

3.a) After $M = 10^4$ and $M = 10^5$ update iterations, do both the "basic" and stochastic Policy Gradient algorithms show convergent behavior to a policy? Which one converges faster? Are the policies obtained using these two different Policy Gradient methods similar? How does N influence the convergence rate of the "basic" Policy Gradient method, and why?

3.b) How does the optimal MDP control policy perform in terms of reward per episode? Perform sufficiently many simulations and describe them quantitatively.

Once an optimal policy has been found, plot the movement of the robot for the initial conditions $z_0 = 1, y_0 \in \{1, 2, 3, 4, 5\}$ and $z_0 = 4, y_0 \in \{2, 3, 5\}$, and at every step, show the probability of the robot performing that specific action.

3.c) Describe qualitatively the robot trajectories, for which the above determined optimal policy has been implemented. Provide arguments as to why this is a desirable or undesirable control policy.

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